

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

Video ID:	e8e36826-f370-4f4a-ae81-caba8fb3c604
Total Frames:	1348
Frame Rate:	25.0 fps (assumed)
Duration:	53.9 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	52	✓
RT-DETR Detections	288	✓
MedSAM2 Detections	303	✓
Consensus (3-Model Agreement)	2	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	2	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 33 frames)

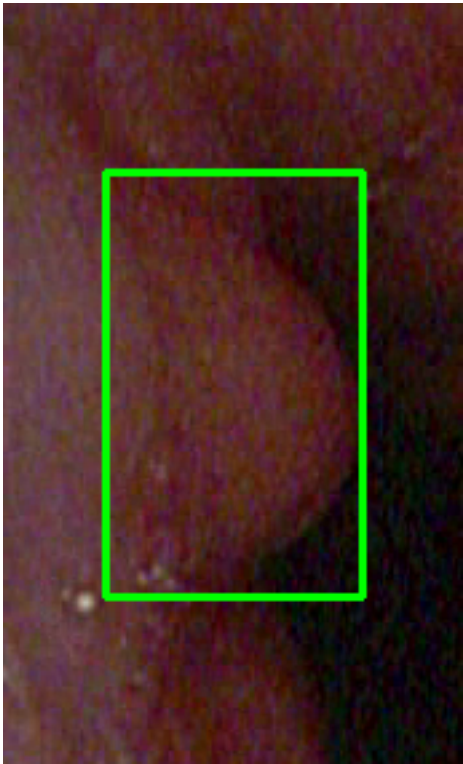


Feature	Value	Interpretation
Frame Index	937	Frame 937 of 1348
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 59.8%
Redness Score	0.058	Color-based vascularization
Relative Radius	7.6%	Polyp size as % of frame diagonal
Texture Score	0.023	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	58.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 28 frames)



Feature	Value	Interpretation
Frame Index	859	Frame 859 of 1348
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_POLYP	Type confidence: 92.0%
Redness Score	0.298	Color-based vascularization
Relative Radius	9.8%	Polyp size as % of frame diagonal
Texture Score	0.058	Surface morphology
Vessel Visibility	0.546	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket

Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	68.3%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 2 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.